

Hospital Readmission Risk Predictor:

A Phased MLOps Build

Exploratory Data Analysis Report

Prepared for: Hospital Administration & Care Management Stakeholders

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GitHub Repository: <https://github.com/AngelineSetiawan/AngelineSetiawan-hospital-readmission-risk-predictor>

July 21, 2026

1. Background & Question

Hospitals across the United States are financially penalized under the Centers for Medicare & Medicaid Services' (CMS) Hospital Readmissions Reduction Program (HRRP) when their 30-day readmission rates exceed expected benchmarks. Roughly one in fourteen eligible hospitals loses at least one percent of Medicare reimbursement under this program, with penalties capped at three percent of payments (Centers for Medicare & Medicaid Services, n.d.). For a hospital administration or care-management team, this translates into a direct, recurring financial exposure that is only partially within the team's control once a patient has already been discharged. The value of a readmission-risk model is that it moves the point of intervention earlier: if a care team can identify which patients are most likely to bounce back within 30 days before those patients ever leave the building, they can target discharge planning, follow-up calls, and medication review at exactly the patients who need it most.

This project's research question is whether a patient's risk of 30-day readmission can be predicted from routinely captured clinical encounter data, and whether that prediction can be delivered through a live, self-updating system rather than a one-time report. This second half of the question matters as much as the first: a model that scores well in a notebook but is never re-evaluated against new data quietly goes stale, and a stale risk score is arguably worse than no score at all, because it creates false confidence. This project therefore treats the predictive model as one component of a larger operational pipeline that logs every experiment, promotes only the best-performing model into production, serves predictions through a live API, and, as a stretch goal, retrains and redeploys itself on a schedule.

The working hypothesis is that patients with more complex or intensive care histories — longer hospital stays, more prior inpatient or emergency visits, and a higher number of medications —

are more likely to be readmitted within 30 days, because these factors reflect greater underlying disease severity and more complicated post-discharge needs. The corresponding prediction is that a supervised classification model will outperform a naive majority-class baseline on standard imbalanced-classification metrics (AUC-ROC and recall for the readmitted class), and that utilization-related features — number of prior inpatient visits, time in hospital, and number of medications — will rank among the model's most important predictors. This exploratory data analysis is a first, direct check on that hypothesis: before any model is trained, does the raw data show these utilization variables moving in the expected direction for patients who were, in fact, readmitted within 30 days? No changes to the research question or hypothesis were made following proposal feedback; the analysis below is the first empirical test of the reasoning laid out in that proposal.

2. Methods

2.1 Dataset

The analysis draws on the Diabetes 130-US Hospitals for Years 1999–2008 dataset, hosted by the UCI Machine Learning Repository (Clore et al., 2014). The dataset spans ten years of clinical care across 130 U.S. hospitals and integrated delivery networks, and it contains 101,766 inpatient diabetic encounters with more than fifty fields per encounter, including demographics, admission and discharge codes, diagnosis codes, medication counts, lab and procedure counts, and utilization in the year prior to each encounter. It is licensed CC BY 4.0 and was obtained as a direct CSV download from the UCI repository (`diabetic_data.csv`), so no API access, credentialing, or web scraping was required.

A second, companion file supplied by UCI, `IDS_mapping.csv`, is not encounter-level data at all — it is a small reference table that stacks three separate ID-to-label lookup tables (one each for

admission type, discharge disposition, and admission source) with a blank line marking the boundary between them. This file was parsed programmatically into three lookup tables and joined onto the main encounter table on the corresponding coded ID columns, producing three new, human-readable label columns (`admission_type_id_label`, `discharge_disposition_id_label`, `admission_source_id_label`) while leaving the original numeric codes untouched for later modeling use. This join is the only data-preparation step involving more than one source file, and it is what makes the categorical summaries in Section 3 legible to a non-technical audience rather than a table of bare integer codes.

All acquisition and preparation code — including the `IDS_mapping` join logic, the data-source attributions and citations, and inline comments explaining each step — is version-controlled in the project's GitHub repository (see title page), together with the data dictionary reproduced in the Appendix (Section A). That repository is structured so another analyst could reproduce every table and figure in this report directly from the original UCI download, rather than from an ad hoc or undocumented notebook.

This dataset was selected over more recently collected alternatives after a deliberate review of what is realistically available. Modern hospital EHR extracts with real readmission outcomes are protected health information under HIPAA and are not released publicly; the closest research-grade modern alternative, MIMIC-IV, requires completed human-subjects research training, a signed data use agreement, and PhysioNet credentialing that could not be secured within this project's timeframe. Synthetic modern alternatives avoid that gatekeeping but do not carry real readmission labels, which would undermine the project's predictive validity. Two limitations of this choice are worth stating plainly for planning purposes. First, the encounters were collected between 1999 and 2008 — before the HRRP existed and before widespread

electronic health record adoption — so any relationships the eventual model learns describe historical patterns in this population and should be read as a proof of concept rather than a claim about current hospital performance. Second, after collapsing the outcome into a binary 30-day label, the positive (readmitted) class shrinks to roughly 11 percent of encounters, which is a meaningful but not enormous sample for model training and will require an evaluation approach that does not rely on raw accuracy.

2.2 EDA Methods

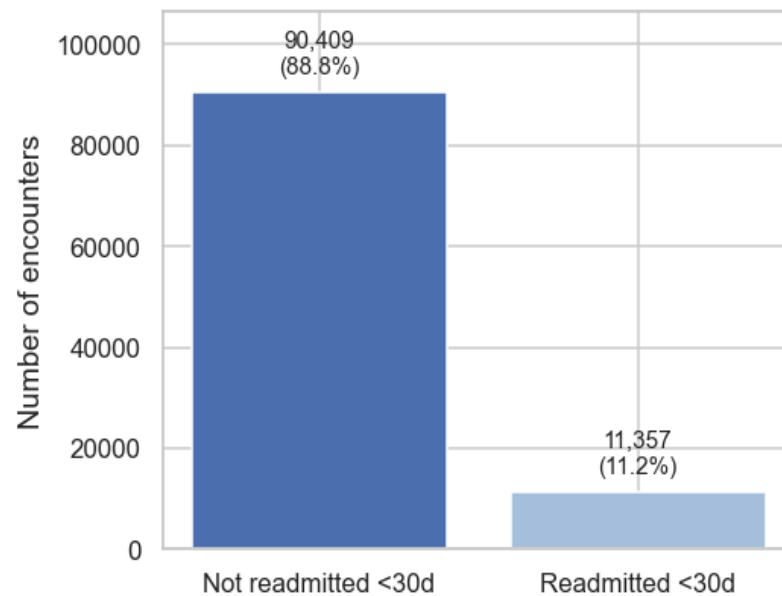
Consistent with the EDA stage of this project, no missing-value imputation and no row deletion were performed at this point. Three narrow, non-destructive steps were taken solely to make the data explorable: the dataset's placeholder "?" values were converted to proper null values so that missingness could be counted accurately; the native three-class readmission label (readmitted in under 30 days, readmitted after 30 days, or not readmitted) was collapsed into the binary target `readmitted_30d` used throughout this report; and one variable, `weight`, was dropped because it is missing for roughly 97 percent of encounters and is not usable in any form at this sample size. No other column was altered, dropped, or imputed.

Descriptive statistics were generated for the eight core continuous predictors most relevant to the hypothesis (length of stay, lab procedures, non-lab procedures, medications, and three prior-utilization counts, plus number of diagnoses) and for a set of key categorical fields spanning demographics, admission and discharge context, test results, and medication activity. For continuous variables this included measures of central tendency and spread (mean, median, standard deviation, minimum, and maximum) as well as skew and excess kurtosis to flag heavily one-sided or long-tailed distributions before they reach a model. For categorical variables this included frequency counts and percentages for every observed category. Visual methods included

histograms to inspect distribution shape, boxplots and a rate-by-bucket line chart to compare hypothesis-relevant predictors directly against the readmission outcome, a bar chart of readmission rate by age bracket to check whether age behaves as expected, and a Pearson correlation heatmap (correlogram) across the continuous predictors and the target to screen for multicollinearity before any feature-selection or regularization decisions are made. A Spearman rank-correlation supplement was also computed against the target to catch monotonic-but-nonlinear relationships that a purely linear Pearson correlation could understate — this matters here because several utilization counts are heavily zero-inflated.

3. Results

The dataset's target variable is highly imbalanced: as Figure 1 shows, only 11.2% of the 101,766 encounters (11,357 encounters) resulted in readmission within 30 days, while the remaining 88.8% did not. This imbalance is the single most important fact shaping the analysis plan going forward — it is the reason evaluation metrics later in this project will lean on AUC-ROC and recall for the positive class rather than raw accuracy, since a model that always predicted "not readmitted" would already be 88.8% accurate while being clinically useless.

Figure 1. Distribution of 30-Day Readmission Outcome**Figure 1. Distribution of the 30-day readmission outcome across all 101,766 encounters.**

Only about one in nine encounters ends in a 30-day readmission. This class imbalance means accuracy alone will be a misleading metric for the eventual classifier, and it is the primary justification for prioritizing AUC-ROC and recall in later model evaluation. Note: this figure is computed on the full, uncleaned sample and still includes the 1,652 encounters ending in patient death (Section 4.2); because those encounters are guaranteed non-readmissions, the true positive rate is expected to shift slightly upward once they are excluded during preprocessing.

3.1 Descriptive Statistics

Table 1 summarizes the eight continuous predictors most relevant to the stated hypothesis. Two patterns stand out. First, the three prior-utilization counts — `number_outpatient`, `number_emergency`, and `number_inpatient` — are extremely right-skewed (skew values of 8.83, 22.86, and 3.61, respectively) with very heavy tails (excess kurtosis in the hundreds to over a thousand for `number_emergency`). In practical terms, most patients have zero prior outpatient, emergency, or inpatient visits in the preceding year, while a small number of patients have dozens. This zero-inflation will need to be accounted for during feature engineering, either through capping, binning, or a model family that tolerates skewed inputs well. Second, `num_medications` and `time_in_hospital` are also right-skewed but far less extreme, which is

consistent with the fact that hospital stays and medication counts are bounded by clinical and administrative limits in a way that outpatient visit counts are not.

Variable	N	Mean	Median	Std. Dev.	Min	Max	Skew	Kurtosis
time_in_hospital	101,766	4.40	4.0	2.99	1	14	1.13	0.85
num_lab_procedures	101,766	43.10	44.0	19.67	1	132	-0.24	-0.25
num_procedures	101,766	1.34	1.0	1.71	0	6	1.32	0.86
num_medications	101,766	16.02	15.0	8.13	1	81	1.33	3.47
number_outpatient	101,766	0.37	0.0	1.27	0	42	8.83	147.90
number_emergency	101,766	0.20	0.0	0.93	0	76	22.86	1,191.63
number_inpatient	101,766	0.64	0.0	1.26	0	21	3.61	20.72
number_diagnoses	101,766	7.42	8.0	1.93	1	16	-0.88	-0.08

Table 1. Descriptive statistics for the eight core continuous predictors ($n = 101,766$).

Table 2 summarizes all twelve categorical variables carried into this EDA — every categorical field named in the analysis plan is represented, covering demographics, admission and discharge context, glucose/HbA1c testing, and diabetes medication activity. For each variable it reports how many categories it takes on, its single most common category, and the missingness or "unknown" rate that applies to it, whether that missingness comes from a true null or from a coded placeholder label such as "NULL" or "Not Available" (see Table 6 below). The full 90-row, category-by-category frequency breakdown that this summary is drawn from — needed to see every individual discharge-disposition or admission-source category, for example — is reproduced in full in the Appendix (Section B) and is also version-controlled in the GitHub repository. Read together, Table 2 shows a patient population that skews older (58.5% of encounters are age 60 or above), is admitted predominantly through emergency or urgent channels (71.2% combined) rather than as scheduled elective care, and is rarely lab-tested for glucose or HbA1c at this encounter (94.8% and 83.3% not tested, respectively) — a pattern more

likely to reflect testing protocol than patient health and worth carrying into the modeling stage as its own signal rather than discarding as "missing."

Variable	# Categories	Most Common Category	% Most Common	% Missing / Unknown
race	6	Caucasian	74.78%	2.23%
gender	3	Female	53.76%	0.00%
age (bracket)	10	[70–80)	25.62%	0.00%
admission_type_id_label	8	Emergency	53.05%	10.22%
discharge_disposition_id_label	26	Discharged to home	59.19%	4.60%
admission_source_id_label	17	Emergency Room	56.50%	6.66%
max_glu_serum	3 + not tested	Not tested	94.75%	0.00%
A1Cresult	3 + not tested	Not tested	83.28%	0.00%
change (med. changed)	2	No change	53.80%	0.00%
diabetesMed (Rx present)	2	Yes	77.00%	0.00%
insulin	4	No	46.56%	0.00%
metformin	4	No	80.36%	0.00%

Table 2. Categorical variable summary — all twelve categorical fields in the analysis plan ($n = 101,766$). Full category-level frequencies are in the Appendix (Section B).

3.2 Hypothesis-Relevant Comparisons

Table 3 stratifies the core continuous predictors by readmission outcome and speaks most directly to the hypothesis. Patients readmitted within 30 days show a higher mean on every one of the eight utilization variables than patients who were not readmitted, and the gap is largest for number_inpatient (mean of 1.12 prior inpatient visits among the readmitted group versus 0.56 among the non-readmitted group — roughly double). Time in hospital, medication count, and emergency visits all move in the same direction, though the differences in means are more modest for those variables. This is an encouraging early signal for the hypothesis, though it should be read cautiously: these are simple two-group mean comparisons, not controlled effect sizes, and the medians for several variables (time_in_hospital, num_procedures) are identical

across groups even where the means differ, which suggests the difference is being driven by the distribution's tail rather than a shift in the typical patient.

Continuous predictor	Not readmitted (mean / median)	Readmitted <30d (mean / median)
time_in_hospital	4.35 / 4.0	4.77 / 4.0
num_lab_procedures	42.95 / 44.0	44.23 / 45.0
num_procedures	1.35 / 1.0	1.28 / 1.0
num_medications	15.91 / 15.0	17.01 / 16.0
number_outpatient	0.36 / 0.0	0.39 / 0.0
number_emergency	0.18 / 0.0	0.36 / 0.0
number_inpatient	0.56 / 0.0	1.12 / 0.0
number_diagnoses	7.35 / 8.0	7.99 / 8.0

Table 3. Continuous predictor means and medians, stratified by 30-day readmission outcome.

Table 4 complements Table 3 by stratifying the same readmission outcome across five of the most policy-relevant categorical predictors, so that both the continuous and categorical sides of the population are checked against the target. Demographic splits (gender, race) show only a narrow 1–2 percentage-point spread in readmission rate across categories, suggesting demographics alone are unlikely to be strong standalone predictors. Treatment-related variables show more separation: patients who were prescribed a diabetes medication were readmitted at 11.63% versus 9.60% for those who were not, and patients whose medication regimen was changed during the encounter were readmitted at 11.82% versus 10.59% for those whose regimen was unchanged. Both gaps point in the same direction as the utilization-based hypothesis in Section 1 — greater treatment intensity or a mid-course medication change tracks with higher readmission risk — and both are plausible additions to the feature set alongside the utilization counts already highlighted in Table 3.

Predictor	Category	30-Day Readmission Rate	N (encounters)
Gender	Female	11.25%	54,708
Gender	Male	11.06%	47,055
Race	Caucasian	11.29%	76,099
Race	African American	11.22%	19,210
Race	Hispanic	10.41%	2,037
Race	Asian	10.14%	641
Race	Other	9.63%	1,506
Diabetes medication prescribed	Yes	11.63%	78,363
Diabetes medication prescribed	No	9.60%	23,403
Medication changed at encounter	Changed	11.82%	47,011
Medication changed at encounter	Not changed	10.59%	54,755
Admission type	Emergency	11.52%	53,990
Admission type	Urgent	11.18%	18,480
Admission type	Elective	10.39%	18,869

Table 4. 30-day readmission rate stratified by key categorical predictors, computed directly from the encounter-level data ($n = 101,766$).

Figure 2. Distributions of Key Continuous Predictors

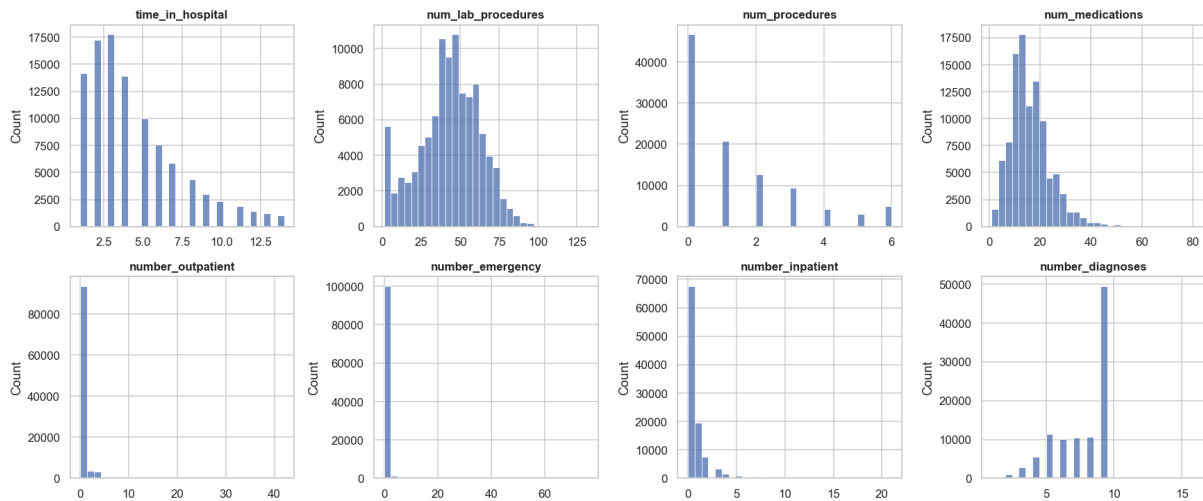


Figure 2. Distributions of the eight core continuous predictors.

Length of stay, lab procedures, and medication count are moderately right-skewed but roughly bell-shaped, while the three prior-utilization counts are dominated by a single spike at zero with a long, thin tail. Number_diagnoses is unusual in the opposite direction — most encounters cluster near the coding maximum of nine diagnoses, which likely reflects a data-entry convention worth flagging during cleaning rather than a true distribution of clinical complexity.

Figure 3. Hypothesis-Relevant Predictors by Readmission Outcome

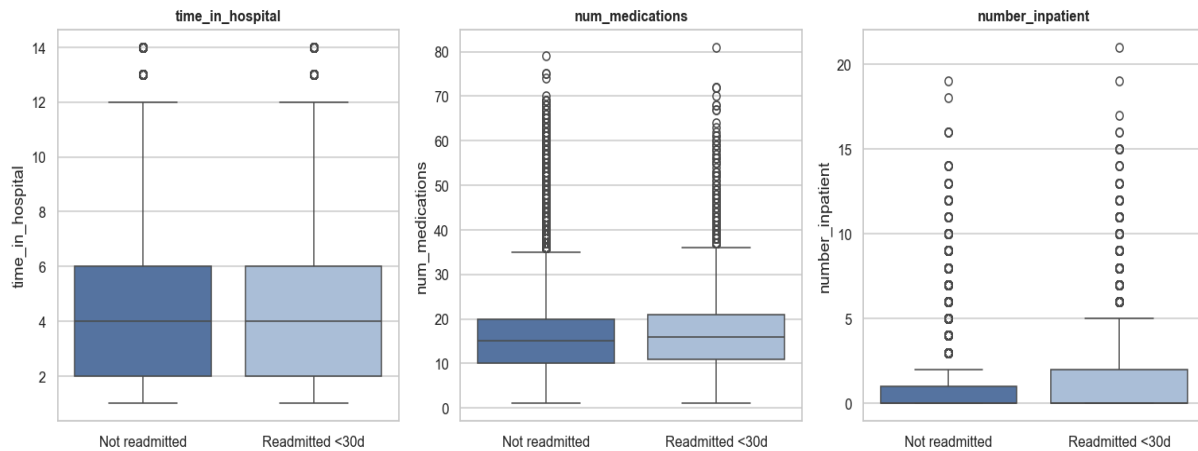


Figure 3. Time in hospital, medication count, and prior inpatient visits, compared by readmission outcome.

The boxplots make the same story visible without averaging it away: the median and interquartile range for `number_inpatient` shift up noticeably for readmitted patients, while `time_in_hospital` and `num_medications` shift only slightly. Prior inpatient utilization currently looks like the strongest of these three hypothesis-relevant predictors. Note: like Figure 1, this is computed prior to excluding the 1,652 structurally-negative expired encounters described in Section 4.2.

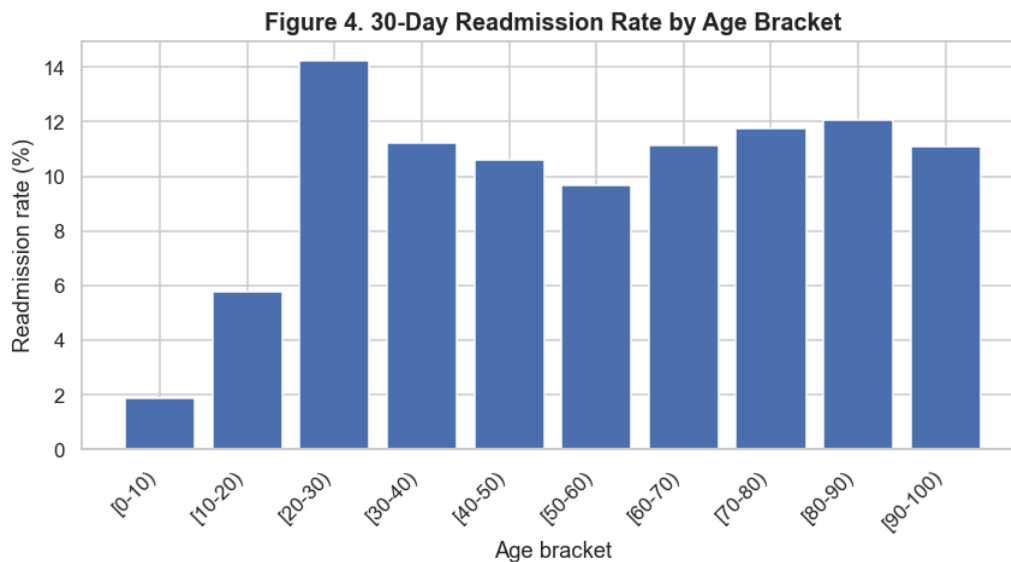


Figure 4. 30-day readmission rate by ten-year age bracket.

Readmission rate rises from under 2% in the youngest bracket to roughly 10–12% across most adult brackets, broadly consistent with age as a proxy for accumulating health burden. The one exception is the [20–30) bracket, where the rate spikes to just over 14%; this bracket is also one of the smallest in the dataset (1,657 encounters, 1.6% of the sample), so the estimate is noisier than the larger adult brackets and is flagged here for a closer look once the data is split for modeling, rather than treated as a confirmed pattern. Note: rates shown here are computed prior

to excluding the 1,652 structurally-negative expired encounters described in Section 4.2, so true age-bracket rates are expected to shift slightly upward, most for the older brackets where in-hospital deaths concentrate.

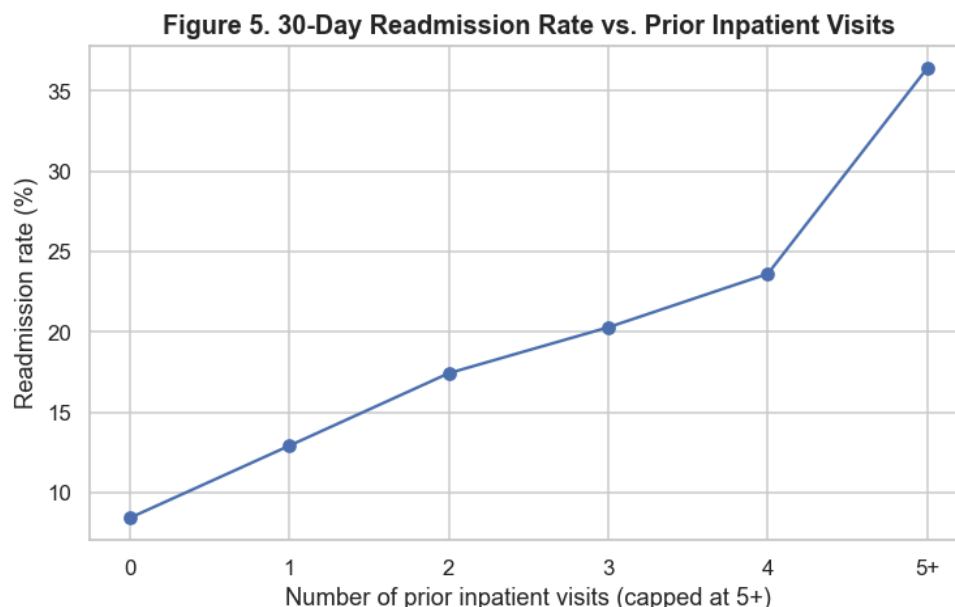


Figure 5. 30-day readmission rate versus number of prior inpatient visits (capped at 5+).

This is the clearest single piece of evidence for the hypothesis in the entire analysis: readmission rate climbs in an almost straight line from about 8% for patients with no prior inpatient visits to over 36% for patients with five or more, a more than four-fold increase. Prior inpatient utilization looks likely to be one of the most important predictors in the eventual model. Note: as in Figures 1, 3, and 4, these rates are computed prior to excluding the 1,652 structurally-negative expired encounters described in Section 4.2.

3.3 Missingness Characterization

Every one of the dataset's 49 columns was inventoried for missingness, not only the handful discussed elsewhere in this report. Under the standardized representation (the dataset's "?" placeholder converted to a true null, for counting purposes only — no values were filled in or removed), nine columns showed any missingness at all, ranging from a negligible 0.02% for `diag_1` up to 96.86% for `weight`; the complete list is in Table 5. The remaining roughly forty analysis-plan columns, including every continuous predictor in Table 1 and the fully-coded categorical fields (`gender`, `age`, `admission_type_id`, `discharge_disposition_id`,

admission_source_id in their raw numeric form), had 0% missingness under this standardized representation.

Column	% Missing	Preliminary handling plan
weight	96.86%	Dropped during EDA (non-informative at this sample size)
max_glu_serum	94.75%	Retain; encode as tested vs. not tested
A1Cresult	83.28%	Retain; encode as tested vs. not tested
medical_specialty	49.08%	Retain; encode missing as its own category
payer_code	39.56%	Retain; encode missing as its own category
race	2.23%	Retain; encode missing as its own category
diag_3	1.40%	Retain; grouped with other diagnosis codes
diag_2	0.35%	Retain; grouped with other diagnosis codes
diag_1	0.02%	Retain; grouped with other diagnosis codes

Table 5. Missingness inventory — all columns with any missing data under the standardized null representation ($n = 101,766$).

A standard null count is not the whole missingness picture for this dataset, however. As Table 6 shows, the three ID-mapping label columns created during data preparation (Section 2.1) carry their own hidden missingness in the form of literal category labels — "NULL," "Not Available," and "Not Mapped" — inherited from the UCI reference table itself. These will not appear in a standard `isna()` count because they are valid strings, not true nulls, so they are quantified separately here to keep the missingness picture complete rather than understated. Between 4.6% and 10.2% of encounters carry one of these hidden-unknown codes, depending on the field, and each will need to be treated as its own explicit "unknown" category during encoding rather than silently folded into a real label.

Coded label column	Count coded "unknown"	% of encounters
admission_type_id_label	10,396	10.22%
discharge_disposition_id_label	4,680	4.60%
admission_source_id_label	6,781	6.66%

Table 6. Hidden missingness — coded "unknown" categories in the three ID-mapping label columns ($n = 101,766$).

3.4 Collinearity Assessment

Figure 6 shows the Pearson correlation matrix across all eight continuous predictors and the target, which satisfies the project's required collinearity check. Correlations among the predictors themselves are modest overall, with the strongest being `time_in_hospital` and `num_medications` ($r = 0.47$) and `num_procedures` and `num_medications` ($r = 0.39$) — both plausible, since longer or more procedure-heavy stays tend to involve more medications. None of the pairwise correlations reach a level (commonly cited thresholds are in the 0.7–0.8 range) that would raise serious multicollinearity concerns for a linear baseline model, though `time_in_hospital`, `num_lab_procedures`, `num_procedures`, and `num_medications` do form a loosely correlated cluster worth monitoring during feature selection.

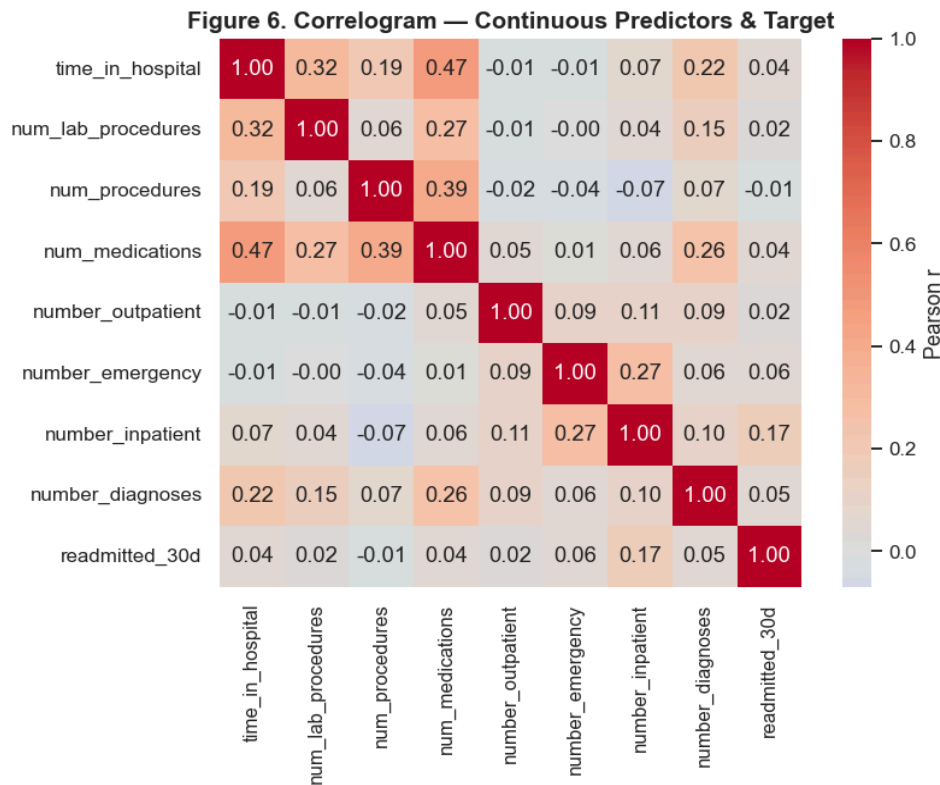


Figure 6. Correlogram of Pearson correlations across continuous predictors and the target (readmitted_30d).

Correlations with the target itself are uniformly weak in linear (Pearson) terms — the strongest is number_inpatient at $r = 0.17$ — which at first looks discouraging for the hypothesis. However, a Spearman rank-correlation check against the target tells a related but distinct story: number_inpatient still leads at 0.138, followed by number_emergency (0.063) and time_in_hospital (0.051), preserving the same relative ordering. Neither correlation coefficient alone captures the step-function relationship visible in Figure 5, because number_inpatient is heavily zero-inflated; this is a reminder that weak linear correlation does not rule out a strong, non-linear relationship, and that tree-based models (random forest, XGBoost) are likely to extract more signal from these utilization variables than a linear correlation coefficient suggests.

4. Discussion & Next Steps

4.1 Key Takeaways

Three findings from this exploratory pass are the most consequential for the project going forward. First, the severe class imbalance documented in Figure 1 (11.2% positive class) confirms that the evaluation plan laid out in the proposal — leaning on AUC-ROC and recall rather than accuracy — is the right call, and reinforces that class-imbalance handling (class weighting, and potentially resampling as a comparison) will need to be a first-class step in preprocessing, not an afterthought. Second, the data lends early, genuine support to the project's hypothesis: readmitted patients show higher utilization on every variable examined in Table 3, and the relationship between prior inpatient visits and readmission rate in Figure 5 is the strongest single pattern in the entire analysis, rising more than four-fold from patients with no prior admissions to patients with five or more. This is an encouraging signal to carry into feature engineering and modeling. Third, the collinearity check in Figure 6 shows no continuous predictor pair correlated highly enough to force an early feature-dropping decision, which keeps the feature set flexible heading into modeling, though the modest weak-linear/stronger-non-linear pattern in number_inpatient (Pearson $r = 0.17$ vs. the step-function shape in Figure 5) is a useful early argument for prioritizing tree-based models over a purely linear baseline.

A fourth takeaway comes from the categorical side of the data (Tables 2 and 4): demographic predictors (gender, race) separate readmission risk only weakly, while treatment-intensity predictors — whether a diabetes medication was prescribed at all, and whether the regimen was changed mid-encounter — separate it more clearly, in the same direction as the utilization hypothesis. This suggests the eventual feature set should lean on treatment and utilization signals rather than demographics as the primary risk drivers, which is also consistent with how the model is intended to be used: care teams act on modifiable clinical signals, not on demographic categories. Finally, the missingness audit in Tables 5 and 6 confirms that missingness in this dataset is not random or evenly spread — it clusters heavily in a handful of lab-result and specialty fields, and several coded fields carry a second, hidden layer of "unknown" values that a naive null count would miss entirely. Because several of the most-missing fields (max_glu_serum, A1Cresult) look more like a testing-protocol signal than noise, the cleaning plan below treats them as informative missingness to be encoded, not simply imputed or dropped.

None of these findings changes the underlying research question or hypothesis stated in Section 1. If anything, the utilization-based hypothesis looks more directly supported by this first look at the data than anticipated, particularly for prior inpatient visits. The one open question this analysis raises rather than answers is the unexpected readmission-rate spike in the youngest adult bracket (Figure 4); because that bracket is small, it will be revisited once a proper train/test split is in place rather than acted on now.

4.2 Data Cleaning & Preprocessing Plan

This EDA surfaced several concrete issues that revise and sharpen the cleaning plan described in the original proposal. None of these have been acted on yet — cleaning is intentionally deferred to the next phase — but each is now a scoped, specific task rather than a general intention.

- Structural leakage from in-hospital deaths: 1,652 encounters (1.62% of the dataset) end in patient death (discharge disposition codes 11, 19, 20, and 21). Every one of these rows is coded "not readmitted" by construction, since a deceased patient cannot be readmitted — meaning these rows are guaranteed negative-class examples for a reason that has nothing to do with clinical risk. Consistent with the original dataset publication (Strack et al., 2014), these encounters will be excluded during cleaning rather than left in as ordinary negative examples, to avoid quietly biasing the model toward an artificially easy negative class.
- High and hidden missingness: Standard missingness (the "?" placeholder) ranges from negligible for diag_1 (0.02%) up to 96.86% for weight, 94.75% for max_glu_serum, and 83.28% for A1Cresult, with medical_specialty (49.08%) and payer_code (39.56%) also substantial. Weight has already been dropped as non-informative at this sample size. The remaining high-missingness fields (max_glu_serum, A1Cresult, medical_specialty, payer_code) will need a deliberate strategy — most likely a "tested/not tested" or "specialty known/unknown" indicator rather than imputing a specific lab value or specialty that was never recorded. Separately, three coded fields (admission_type, discharge_disposition, admission_source) contain hidden missingness in the form of literal category labels such as "NULL," "Not Available," and "Not Mapped," affecting 10.22%, 4.60%, and 6.66% of encounters respectively. These will not surface in a standard null-count and must be handled explicitly as their own missing/unknown category rather than treated as informative labels.
- High-cardinality diagnosis codes: diag_1, diag_2, and diag_3 are ICD-9 codes with hundreds of distinct raw values. As planned in the proposal, these will be grouped into a smaller number of clinically meaningful categories before encoding rather than one-hot encoded directly, to avoid a feature space that is large relative to the number of available rows.
- Class imbalance: With the positive class at roughly 11%, preprocessing will apply class weighting as the primary approach, with resampling methods such as SMOTE evaluated as a comparison rather than a default, consistent with the proposal.
- Encoding and leakage guardrails: Remaining categorical fields (admission type, discharge disposition, admission source, age bracket) will be encoded for modeling, and any field only known after discharge will be explicitly dropped or masked so it cannot leak post-outcome information into the predictors.
- Time-aware, patient-grouped validation: Because the public release does not include exact calendar dates or hospital identifiers, encounter_id will continue to serve as an ordinal proxy for chronological order, with the earliest ~75–80% of encounters forming the training set and the most recent ~20–25% held out as a test set. Splits will additionally be grouped by patient_nbr so that no single patient's encounters appear in

both partitions. This design choice is unchanged by the EDA and remains necessary given the dataset's structural limitations.

Taken together, these steps update the preprocessing plan from a general list of intentions into a checklist with specific columns, specific row counts, and specific justifications, which should make the next phase of this project — feature engineering and model training — considerably faster to execute correctly the first time.

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<https://doi.org/10.1155/2014/781670>

Appendix

A. Data Dictionary

The table below lists the variables referenced in this report and their role in the analysis.

Engineered variables are marked as such; they do not exist in the raw UCI download and are constructed during preprocessing to support the time-aware validation strategy described in Section 4.2. readmitted_30d is the target variable for the eventual supervised classifier.

Variable	Description	Data Type	Role
encounter_id	Unique identifier for a single inpatient encounter.	Integer (ID)	Row key
patient_nbr	Unique patient identifier; repeats across a patient's multiple encounters.	Integer (ID)	Grouping variable for patient-level train/test split
race	Patient's recorded race/ethnicity category.	Categorical	Predictor
gender	Patient's recorded gender.	Categorical	Predictor
age	Ten-year age bracket (e.g., [50-60)).	Ordinal categorical	Predictor
admission_type_id	Coded category describing how the admission occurred (e.g., emergency, elective, urgent).	Categorical (coded)	Predictor
discharge_disposition_id	Coded category describing discharge destination (e.g., home, transferred, hospice, expired).	Categorical (coded)	Predictor
admission_source_id	Coded category describing the referral source for admission.	Categorical (coded)	Predictor
time_in_hospital	Length of stay in days for the encounter.	Integer	Predictor (core to hypothesis)
num_lab_procedures	Number of lab tests performed during the encounter.	Integer	Predictor
num_procedures	Number of non-lab procedures performed during the encounter.	Integer	Predictor
num_medications	Number of distinct medications administered during the encounter.	Integer	Predictor (core to hypothesis)
number_outpatient	Number of outpatient visits in the year preceding the encounter (already lagged).	Integer	Predictor (core to hypothesis)
number_emergency	Number of emergency visits in the year preceding the encounter (already lagged).	Integer	Predictor (core to hypothesis)
number_inpatient	Number of inpatient visits in the year preceding the encounter (already lagged).	Integer	Predictor (core to hypothesis)
diag_1 / diag_2 / diag_3	Primary, secondary, and tertiary ICD-9 diagnosis codes (high-cardinality).	Categorical (coded)	Predictor (to be grouped/encoded)

Variable	Description	Data Type	Role
number_diagnoses	Total number of diagnoses recorded for the encounter.	Integer	Predictor
max_glu_serum / A1Cresult	Glucose serum and HbA1c test results, if performed.	Categorical	Predictor
change / diabetesMed	Whether diabetic medication was changed, and whether any diabetic medication was prescribed.	Binary categorical	Predictor
weight	Patient weight; ~97% missing. Dropped during EDA as non-informative.	Categorical	Dropped
encounter_order (engineered)	Ordinal rank of an encounter based on encounter_id, used as a chronology proxy.	Integer (engineered)	Time-aware split key
days_since_prior_encounter (engineered)	Gap, in encounter order, since the same patient's previous recorded encounter.	Integer (engineered)	Predictor (lag/elapsed-time feature)
readmitted	Native three-class label: readmitted in <30 days, readmitted in >30 days, or not readmitted.	Categorical	Source of target variable
readmitted_30d (engineered)	Binary collapse of readmitted: 1 if readmitted within 30 days, 0 otherwise.	Binary integer (0/1)	TARGET

B. Full Categorical Variable Frequency Table

This table is the complete, category-by-category expansion that Table 2 summarizes. It covers all twelve categorical variables in the analysis plan and every observed category within each, and is provided for readers who want the full distribution behind a specific field (for example, every individual discharge-disposition code) rather than just its most common category. This table, and the code that generated it, is version-controlled in the project's GitHub repository alongside the raw data.

Variable	Category	Count	% of Encounters
race	Caucasian	76,099	74.78%
race	AfricanAmerican	19,210	18.88%
race	(missing / null)	2,273	2.23%
race	Hispanic	2,037	2.00%
race	Other	1,506	1.48%

Variable	Category	Count	% of Encounters
race	Asian	641	0.63%
gender	Female	54,708	53.76%
gender	Male	47,055	46.24%
gender	Unknown/Invalid	3	0.00%
age	[70-80)	26,068	25.62%
age	[60-70)	22,483	22.09%
age	[50-60)	17,256	16.96%
age	[80-90)	17,197	16.90%
age	[40-50)	9,685	9.52%
age	[30-40)	3,775	3.71%
age	[90-100)	2,793	2.74%
age	[20-30)	1,657	1.63%
age	[10-20)	691	0.68%
age	[0-10)	161	0.16%
admission_type_id_label	Emergency	53,990	53.05%
admission_type_id_label	Elective	18,869	18.54%
admission_type_id_label	Urgent	18,480	18.16%
admission_type_id_label	(missing / null)	5,291	5.20%
admission_type_id_label	Not Available	4,785	4.70%
admission_type_id_label	Not Mapped	320	0.31%
admission_type_id_label	Trauma Center	21	0.02%
admission_type_id_label	Newborn	10	0.01%
discharge_disposition_id_label	Discharged to home	60,234	59.19%
discharge_disposition_id_label	Discharged/transferred to SNF	13,954	13.71%
discharge_disposition_id_label	Discharged/transferred to home with home health service	12,902	12.68%
discharge_disposition_id_label	(missing / null)	3,691	3.63%
discharge_disposition_id_label	Discharged/transferred to another short term hospital	2,128	2.09%
discharge_disposition_id_label	Discharged/transferred to another rehab facility (incl. hospital rehab units)	1,993	1.96%
discharge_disposition_id_label	Expired	1,642	1.61%
discharge_disposition_id_label	Discharged/transferred to another type of inpatient care institution	1,184	1.16%
discharge_disposition_id_label	Not Mapped	989	0.97%

Variable	Category	Count	% of Encounters
discharge_disposition_id_label	Discharged/transferred to ICF	815	0.80%
discharge_disposition_id_label	Left AMA	623	0.61%
discharge_disposition_id_label	Discharged/transferred to a long term care hospital	412	0.40%
discharge_disposition_id_label	Hospice / home	399	0.39%
discharge_disposition_id_label	Hospice / medical facility	372	0.37%
discharge_disposition_id_label	Discharged/transferred/referred to a psychiatric hospital or unit	139	0.14%
discharge_disposition_id_label	Discharged/transferred to home under care of Home IV provider	108	0.11%
discharge_disposition_id_label	Discharged/transferred within institution to Medicare-approved swing bed	63	0.06%
discharge_disposition_id_label	Discharged/transferred to a nursing facility (Medicaid only, not Medicare)	48	0.05%
discharge_disposition_id_label	Admitted as an inpatient to this hospital	21	0.02%
discharge_disposition_id_label	Discharged/transferred/referred to this institution for outpatient services	14	0.01%
discharge_disposition_id_label	Discharged/transferred/referred to another institution for outpatient services	11	0.01%
discharge_disposition_id_label	Expired at home; Medicaid only, hospice	8	0.01%
discharge_disposition_id_label	Neonate discharged to another hospital for neonatal aftercare	6	0.01%
discharge_disposition_id_label	Discharged/transferred to a federal health care facility	5	0.00%
discharge_disposition_id_label	Still patient, or expected to return for outpatient services	3	0.00%
discharge_disposition_id_label	Expired in a medical facility; Medicaid only, hospice	2	0.00%
admission_source_id_label	Emergency Room	57,494	56.50%
admission_source_id_label	Physician Referral	29,565	29.05%
admission_source_id_label	(missing / null)	6,781	6.66%
admission_source_id_label	Transfer from a hospital	3,187	3.13%
admission_source_id_label	Transfer from another health care facility	2,264	2.22%
admission_source_id_label	Clinic Referral	1,104	1.08%
admission_source_id_label	Transfer from a Skilled Nursing Facility (SNF)	855	0.84%
admission_source_id_label	HMO Referral	187	0.18%

Variable	Category	Count	% of Encounters
admission_source_id_label	Not Mapped	161	0.16%
admission_source_id_label	Not Available	125	0.12%
admission_source_id_label	Court/Law Enforcement	16	0.02%
admission_source_id_label	Transfer from hospital inpt/same facility, separate claim	12	0.01%
admission_source_id_label	Transfer from critical access hospital	8	0.01%
admission_source_id_label	Extramural Birth	2	0.00%
admission_source_id_label	Normal Delivery	2	0.00%
admission_source_id_label	Transfer from Ambulatory Surgery Center	2	0.00%
admission_source_id_label	Sick Baby	1	0.00%
max_glu_serum	(missing / null — not tested)	96,420	94.75%
max_glu_serum	Norm	2,597	2.55%
max_glu_serum	>200	1,485	1.46%
max_glu_serum	>300	1,264	1.24%
A1Cresult	(missing / null — not tested)	84,748	83.28%
A1Cresult	>8	8,216	8.07%
A1Cresult	Norm	4,990	4.90%
A1Cresult	>7	3,812	3.75%
change	No	54,755	53.80%
change	Ch (changed)	47,011	46.20%
diabetesMed	Yes	78,363	77.00%
diabetesMed	No	23,403	23.00%
insulin	No	47,383	46.56%
insulin	Steady	30,849	30.31%
insulin	Down	12,218	12.01%
insulin	Up	11,316	11.12%
metformin	No	81,778	80.36%
metformin	Steady	18,346	18.03%
metformin	Up	1,067	1.05%
metformin	Down	575	0.57%

Table B1. Complete categorical variable frequency breakdown, all 90 observed (variable, category) pairs ($n = 101,766$).